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Research Article

Unravelling dengue serotype 3 transmission in Brazil: evidence for multiple introductions of the 3III_B.3.2 lineage

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Abstract

Dengue, caused by dengue virus (DENV) 1–4, remains a global public health concern, with Brazil experiencing some of the largest epidemics. The re-emergence of DENV-3 in Brazil between 2023 and 2024 has raised concerns about new outbreaks due to the absence of sustained circulation of this serotype in recent years. This study investigates the dynamics of DENV-3 in Brazil, focusing on the spread of the 3III_B.3.2 lineage within genotype 3III and its introduction routes. We analysed 1536 DENV-3 genomes, including 11 newly generated in this study, all classified as genotype 3III, the dominant DENV-3 genotype in Brazil since 2001. Phylogenetic analysis identified the 3III_B.3.2 lineage in all recent Brazilian cases, with detections also reported in Central America, the USA, and Europe. At least six independent introduction events of this lineage into Brazil were identified, with the Caribbean region as the primary source. The earliest introduction likely occurred in late 2022 in Roraima, followed by introductions in São Paulo, Minas Gerais, and Pará. While one instance of interstate transmission was detected—from São Paulo to Minas Gerais—our findings indicate that external introductions, rather than domestic spread, were the primary drivers of DENV-3 circulation during this period. These results underscore the importance of continued genomic surveillance and coordinated public health strategies to monitor and mitigate future outbreaks.

Keywords: dengue lineage; phylogeography; outbreak; genomic surveillance

Introduction

Dengue fever, the most prevalent arbovirus disease worldwide, is caused by dengue virus (DENV), a member of the *Flaviviridae* family (genus *Orthoflavivirus*, *Orthoflavivirus denguei*, DENV) (Postler et al. 2023). The main transmission route for DENV is through the bites of female mosquitoes from the *Aedes* genus, particularly *Aedes aegypti* and *Aedes albopictus*. However, other transmission routes, such as blood transfusion (Linnen et al. 2008, Sabino et al. 2016) and vertical transmission (Zambrano et al. 2024) have also been observed. Clinically, the infection ranges from mild to severe, with the latter manifesting as hypovolemic shock due to plasma leakage, which is also associated with high mortality. The reason why some patients develop severe forms of DENV is not fully understood, but the presence of heterologic antibodies to other subtypes has been suggested as a primary factor (Halstead 1988, St. John and Rathore 2019).

DENV is classified into four distinct serotypes (DENV-1 to DENV-4), which are phylogenetically related (Rico-Hesse 1990). Moreover, within each serotype, there is genetic variability, with multiple genotypes identified. Brazil is hyperendemic for DENV, with the circulation of all four known serotypes. However, recent outbreaks have been mainly caused by serotypes 1 and 2, with sporadic reports of the other two serotypes (Ministério da Saúde 2024). Additionally, the country is the global leader in dengue cases, with 7 191 748 confirmed cases between 2014 and 2023 (Pan American Health Organization 2025).

In 2024, more than 10 million cases were reported, of which 57.61% were confirmed, accompanied by a significant increase in mortality, with 6161 deaths (Pan American Health Organization 2025). Although dengue virus type 3 (DENV-3) accounted for a small fraction of these cases, its incidence rose sharply, with 1694 reported cases in Brazil—a significant increase compared to the 106 cases recorded in 2023. This surge was particularly notable in the Northern and Southeastern regions, where detection rates were the highest (Ministério da Saúde 2025, 2024).

DENV-3 is categorized into five distinct genotypes (previously referred to as I–V), which have been updated based on the new lineage nomenclature, with the genotypes now designated as 3I, 3II, 3III, 3IV, and 3V (Hill et al. 2024). Among them, genotype 3III is the most widespread and associated with major outbreaks in Asia, Africa, and the Americas (Messer et al. 2003). In Brazil, the first autochthonous transmission of this genotype was reported in the late 2000s, marking the introduction of this serotype in the country (Miagostovich et al. 2002). Recent reports have indicated a notable increase in the detection of DENV-3, suggesting a resurgence of this serotype in frequency and geographic distribution (Naveca et al. 2023; Adelino et al. 2024; Taylor-Salmon et al. 2024), associated with the emergence and replacement of lineages over the years (Supplementary Fig. 1).

This situation poses a significant risk for the emergence of new outbreaks in Brazil, particularly given the historical absence of sustained DENV-3 circulation and the probable low serological coverage. In this context, our study investigates the recent spread of DENV-3 in Brazil using a phylogeographic approach, with a focus on the classification of circulating lineages and the detection of 11 new cases in the Northern and Southeastern regions of the country.

Materials and methods

Clinical sample collection and genome sequencing

The CeVIVAS initiative

The samples used in this study were provided by the CeVIVAS (Center for Viral Surveillance and Serological Assessment), a collaborative initiative spearheaded by Instituto Butantan in partnership with Central Public Health Laboratories (LACEN—from the Portuguese *Laboratório Central de Saúde Pública*) and other institutions. CeVIVAS aims to strengthen viral genomic surveillance and serological assessment efforts in the country (https://bv.fapesp.br/en/auxilios/110575/continuous-improvement-of-vaccines-center-for-viral-surveillance-and-serological-assessment-cevivas/enter-for-viral-surveillance-and-serological-assessment-cevivas/). This study was approved by the Institutional Ethics Committee of the Faculty of Medicine of the University of São Paulo (Decision number CAAE: 68586623.0.0000.0068).

Clinical samples

The samples analysed in this study were collected between epidemiological weeks 4 and 44 at LACEN Pará ($n = 2$) and LACEN Minas Gerais ($n = 9$) and were confirmed as DENV-3, using the CDC-established protocol (Santiago et al. 2013, Centers for Disease Control 2025), exhibiting Ct values below 30. The RNA extraction was performed to all samples using the Extracta kit AN viral (Loccus) in an automated extractor (EXTRACTA 96, Loccus) according to the manufacturers' specifications. The viral RNA was confirmed using multiplex TaqMan real-time PCR (GeneFinderTM DENV Typing RealAmp Kit, OSang Healthcare Co. Ltd.), which detects all four serotypes and targets the 3´UTR genomic portion. DENV sequencing was performed using primer sets (Supplementary Table 1) specific to DENV-3. Genomic libraries were prepared using the COVIDSeq protocol, following the manufacturer's instructions, except for the used primers. The pooled libraries were normalized to 4 nM, denatured with 0·2 N NaOH and 400 mM Tris–HCl (pH 8), and sequenced using the NextSeq 500/550 High Output Kit v2.5 (300 cycles) on the Illumina NextSeq 2000.

Genome assembly pipeline

The assembly pipeline employed, is part of the VIPER (Viral Identification Pipeline for Emergency Response), which is publicly accessible at https://github.com/V-GEN-Lab/viper. Comprehensive descriptions of the specific parameters and software utilized are provided in Supplementary Material and detailed in Supplementary Fig. 2.

Classification of DENV-3 genomes and phylogenomic reconstruction

We selected all complete DENV-3 genomes (genome size >10 kb) collected from human hosts and available in GenBank ($n = 2347$), using the NCBI Virus portal (https://www.ncbi.nlm.nih.gov/labs/virus/vssi/#/), accessed on 22 August 2024. Additionally, we retrieved all DENV-3 Brazilian complete genomes (>10 kb) collected from human hosts and available in GISAID EpiArbo (Wallau et al. 2023), collected from 1 January 2022, onward, with complete collection date information ($n = 26$). The dataset (EPI_SET_250604wy) (https://doi.org/10.55876/gis8.250604wy) was restricted to sequences associated with publications or deposited by laboratories collaborating with the CeVIVAS project (see Acknowledgements). This dataset, along with the 11 genomes generated in this study (final dataset $n = 2387$), was classified using Nextclade (Aksamentov et al. 2021), with the DENV-3 reference dataset, following the new Dengue Lineages nomenclature (Hill et al. 2024). All genomes belonging to genotype 3III were selected, as all Brazilian sequences belonged to this genotype.

The dataset of DENV-3 genotype 3III ($n = 1536$; metadata available at Zenodo https://doi.org/10.5281/zenodo.13885992 in Supplementary File 1) was aligned using the Augur align tool from the Augur program toolkit v24.1.0 (Huddleston et al. 2021), with the MAFFT v7.520 algorithm (Katoh and Standley 2013). A maximum likelihood (ML) phylogenetic tree was then constructed using IQ-TREE v2.2.6, applying the GTR+F+I+G4 nucleotide substitution model, as determined by ModelFinder within IQ-TREE2. The robustness of the tree topology was evaluated through 10 000 UFBoot replicates.

Spatiotemporal and viral recent spread reconstruction of DENV-3

To investigate the recent spread of DENV-3 in Brazil, we employed a discrete Bayesian phylogeographic inference approach using BEAST v1.10.4 (Suchard et al. 2018). To optimize computational efficiency, branches containing recent Brazilian samples (collected from 2022 onwards) were identified from the ML phylogenetic tree, and the corresponding sequences were retrieved ($n = 88$). Outlier sequences ($n = 4$), with temporal values deviating by more than 1.5 interquartile ranges in root-to-tip regression analysis, identified using TempEst v1.5.3 (Rambaut et al. 2016), were excluded (Naveca et al. 2023). Time-scaled phylogenetic trees were inferred using a relaxed molecular clock model (Drummond et al. 2006). Ancestral states were reconstructed based on a continuous-time Markov chain prior (Ferreira and Suchard 2008) with discrete spatial diffusion, using an asymmetric substitution model to infer migration events (Lemey et al. 2009). Bayesian stochastic search variable selection was implemented to identify statistically supported migration routes. Two independent Markov chain Monte Carlo (MCMC) chains were run for 200 million generations each. Convergence was evaluated by calculating the effective sample size (ESS > 200) for all parameters using Tracer v1.7.1 (Rambaut et al. 2018). The log files are available at Zenodo https://doi.org/10.5281/zenodo.13885992 in Supplementary Files 2 and 3. After discarding the initial 10% as burn-in, the maximum clade credibility (MCC) tree was summarized with TreeAnnotator (Rambaut and Drummond 2013). The final time-scaled phylogeny was visualized in FigTree v1.4.5 (Rambaut 2009), and spatiotemporal information embedded in the posterior distribution of trees was extracted and mapped using the R package 'seraphim' (version 1.0) (Dellicour et al. 2016). The metadata for the final dataset is available in Supplementary Table 2.

Results

The 11 genomes generated in this study (GISAID EPI_SET_250604dg) were obtained from samples collected between epidemiological weeks 4 and 44 of 2024. The samples, confirmed as DENV-3 (see Methods for details), exhibited Ct values below 30, with genomic coverages ranging from 65.7% to 91.7%, resulting in an overall mean coverage of 79.62%.

Of the 2 387 DENV-3 sequences analysed, 1536 were classified as genotype 3III, the only genotype detected among the 117 sequences from Brazil, except for a single sample collected in 2006 (GenBank accession OQ727062). Our analysis reclassified this sample as genotype V, as it was originally labelled as genotype I in the GenBank database. Among the three major lineages within genotype 3III, only 3III_B and 3III_C were identified in Brazil, each circulating during distinct periods (Fig. 1A). Lineage 3III_C was detected exclusively until 2009, whereas lineage 3III_B, specifically minor lineage 3III_B.3.2, has been associated with all Brazilian genomes collected since 2023 ($n = 39$), including the 11 genomes sequenced in this study.

In 2023, Brazil recorded 1 417 218 confirmed DENV cases. Although only 106 of these were attributed to DENV-3 infections, this number represents a significant increase compared to the previous year, when only four DENV-3 cases were confirmed out of 1 210 760 total DENV cases in the country (Ministério da Saúde 2025, Pan American Health Organization 2025). This upward trend continued into 2024, with 1694 DENV-3 cases reported (Fig. 1B) (Ministério da Saúde 2025). São Paulo recorded the highest number of cases, reporting 510 probable infections, followed by Minas Gerais with 302 cases. Among the 22 Brazilian states listed in the case notification system, 17 reported at least one DENV-3 case in 2024.

Phylodynamic analyses of genotype 3III sequences, particularly within minor lineage 3III_B.3.2, revealed a close evolutionary relationship between Brazilian sequences and those from the Caribbean region, including two sequences from Italy (Supplementary Fig. 3), as previously suggested (Branda et al. 2023, Miguel et al. 2024). To investigate the dispersion patterns of the 3III_B.3.2 lineage in Brazil and its potential introduction routes, we performed a phylogeographic analysis incorporating all Brazilian samples alongside the most closely related sequences from Cuba, USA, Costa Rica, Dominican Republic, Italy, Puerto Rico, Suriname, and Haiti. This dataset ($n = 84$) of 3III_B.3.2 lineage sequences had a strong temporal signal (Fig. 2A) and provided evidence of multiple independent introductions of this lineage into Brazil. The earliest introduction event probably occurred in October 2022 (Figs 2C and 3), likely originating from Cuba and entering the state of Roraima, in Northern Brazil. Additionally, six other introduction events were identified across several Brazilian states from the Caribbean region. The most recent introduction of the 3III_B.3.2 lineage into the state of Pará was estimated to have occurred in February 2024, originating from Cuba.

Two sequences from the state of São Paulo, Southeast Brazil, with collection dates of 25 December 2023 and 5 May 2024, were located on distinct branches of the MCC tree, each tracing back to different ancestral nodes from Cuba and Puerto Rico. However, this inference remains uncertain due to the low posterior probability of these nodes in the phylogenetic tree. We also identified one probable internal dispersal event of the 3III_B.3.2 lineage within Brazil: from São Paulo to Minas Gerais (Southeast) around mid-May 2023.

Discussion

The 2023/2024 dengue epidemic in Brazil stands as one of the most severe in recent decades, with over 10 million reported cases and more than 5 million confirmed cases (Pan American Health Organization 2025), predominantly driven by serotypes 1 and 2 (Ministério da Saúde 2024). However, the re-emergence of DENV-3, after more than 13 years of limited circulation, introduced a new dynamic to the epidemic. The prolonged absence of serotype 3 likely left a substantial portion of the population seronegative for DENV-3, heightening the risk of large-scale outbreaks similar to those witnessed in 2002 (Nogueira et al. 2005, de Araújo et al. 2012).

From 2014 to 2022, only 109 DENV-3 cases were reported in Brazil, with the peak occurring in 2016 (42 cases), as documented in the Brazilian Notifiable Diseases Information System (SINAN) database (see Supplementary File 4). This trend shifted dramatically in 2023, with cases surging to 106, signifying the virus's resurgence, with detections in Roraima (Northern), Paraná (Southern), and Minas Gerais (Southeast) (Naveca et al. 2023, Adelino et al. 2024).

Historically, DENV-3 sequences from Brazil up until 2009 were primarily classified as belonging to the major lineage 3III_C. Within this lineage, the predominant subgroup was the minor lineage 3III_C.2, except for the detection of a single sequence belonging to the minor lineage 3III_C.1 in 2003, specifically in Northern Brazil. Moreover, among the 77 genomes identified until 2009, we observed an increase in the number of available genomes between 2006 and 2007, reflecting the characterization of the Brazilian minor lineage 3III_C.2.2 (identified as clade A) by Villabona-Arenas et al. (2013) in São José do Rio Preto, São Paulo, during the DENV outbreaks between 2006 and 2008, when confirmed DENV cases surpassed 1 million (Mondini et al. 2009, Pan American Health Organization 2025).

Genomic surveillance revealed lineage 3III_B.3.2 as the dominant lineage of genotype 3III worldwide since its first detection (Supplementary Fig. 1) and the complete replacement of the circulating DENV-3 viral population in Brazil between 2023 and 2024. This shift may be associated with multiple factors, including

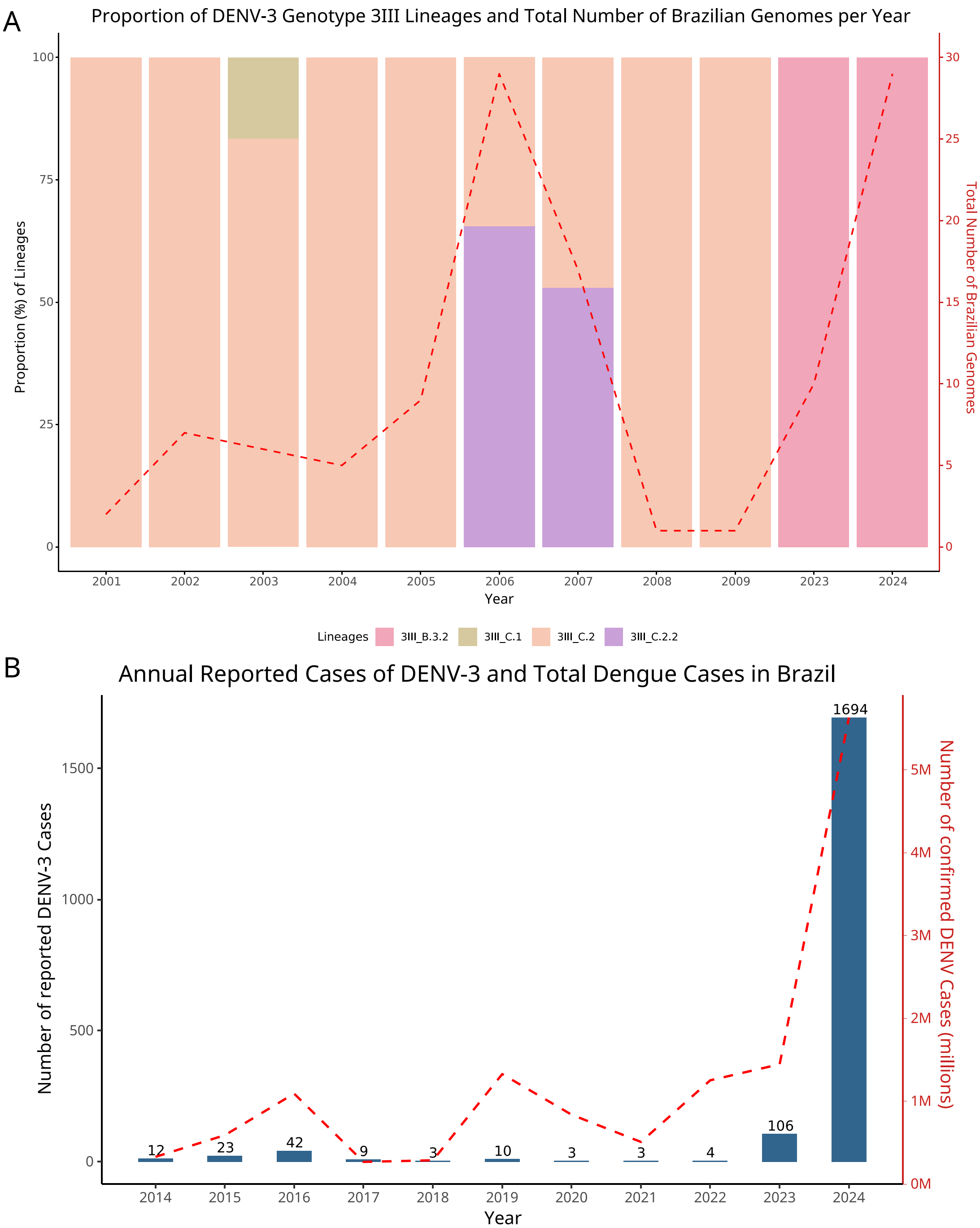

Figure 1. (A) Proportion of DENV-3 genotype 3III lineages and total number of Brazilian genomes per year ($n = 117$). The left Y-axis shows the percentage of each lineage relative to the total number of sequenced samples per year (X-axis). The colours represent specific lineages: 3III_B.3.2, 3III_C.1, 3III_C.2, and 3III_C.2.2. The right Y-axis displays the total number of Brazilian genomes sequenced each year, represented by the dashed line. The absence of bars for certain years (2010–2022) reflects a lack of available genomic data for these periods. (B) Annual Reported Cases of DENV-3 and Total Dengue Cases in Brazil. The Y-axis on the left indicates the number of reported DENV-3 cases, while the X-axis represents the years. The dashed line represents the total number of confirmed dengue cases in Brazil, which corresponds to the secondary Y-axis on the right. Data sourced from PAHO (2025) and the Brazilian Notifiable Diseases Information System—SINAN [TabNet, http://tabnet.datasus.gov.br/cgi/deftohtm.exe?sinannet/cnv/denguebbr.def], accessed on 25 February 2025, and available on Zenodo at https://doi.org/10.5281/zenodo.13885992 in Supplementary File 4.

human mobility, ecological aspects of the vector, and population immunity dynamics.

Air travel between Brazil and the Caribbean region, identified as the primary source of this lineage's introduction into the country, suggests a central role of traveller movement in its dissemination, a phenomenon also observed in other countries in the Americas (Taylor-Salmon et al. 2024). Additionally, the prolonged absence of sustained serotype 3 circulation, even in historically endemic regions for DENV (Ministério da Saúde 2025), may have created a favourable environment for this lineage, possibly due to competitive advantages in transmission compared to those recorded until 2009 (Taylor-Salmon et al. 2024). Environmental factors, such as the territorial expansion of the vector (Codeco et al. 2022, de Oliveira et al. 2023), particularly when combined

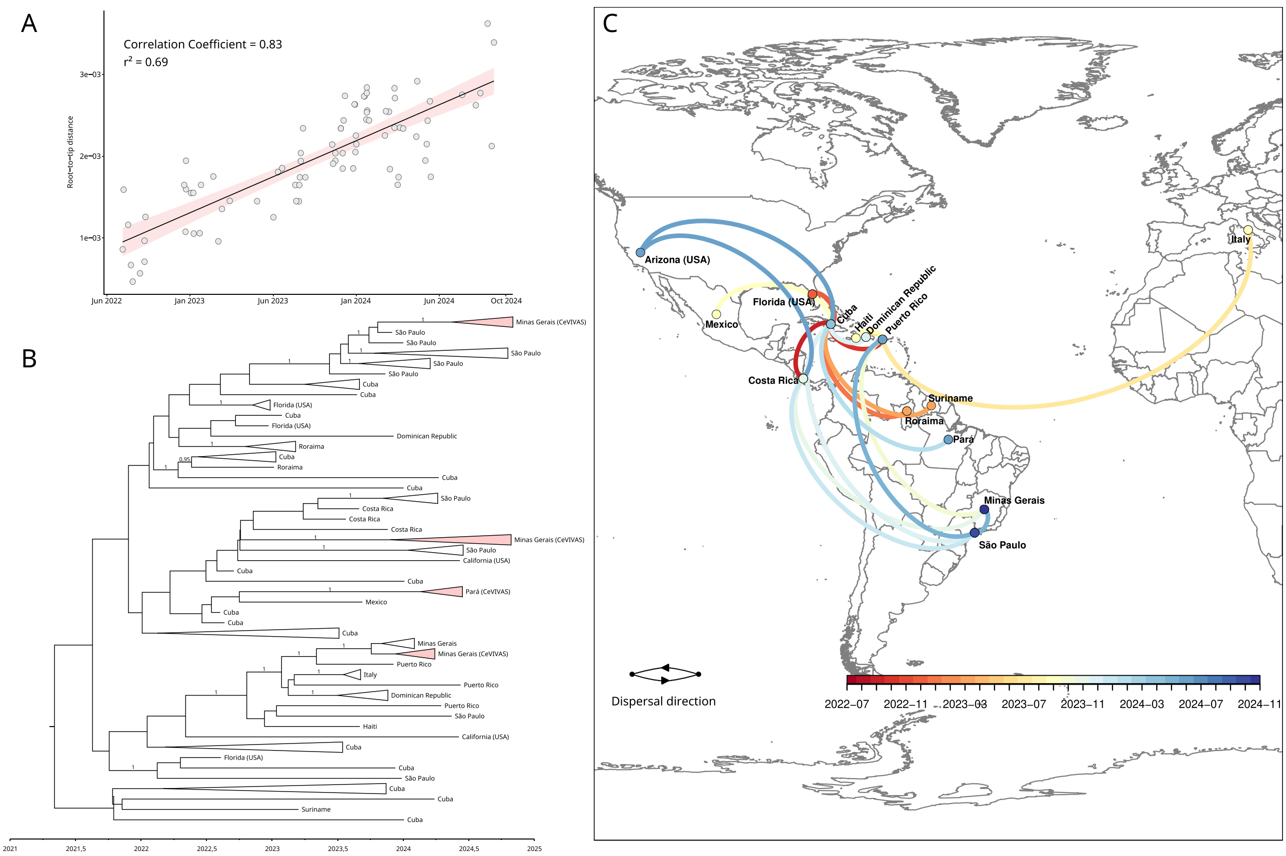

Figure 2. Phylogenetic analysis and geographical distribution of the 3III_B.3.2 lineage from 2021 to 2024. (A) Linear regression between genetic distance and time (in years) shows a significant correlation (correlation coefficient = 0.83, r^2 = 0.69), indicating the temporal evolution of the virus. (B) MCC tree based on genomic sequences from Brazil, with the addition of sequences from Cuba, USA, Dominican Republic, Costa Rica, Mexico, Puerto Rico, Italy, Haiti, and Suriname. The posterior probability values for branches associated with introduction and dispersal events in Brazil are also indicated on the tree. The shaded triangle highlights the genomes generated in this study. (C) Geographic map showing the spread of the analysed sequences. The arrows indicate the dispersal direction of the virus over time, from 2022 to 2024. A dynamic geographic map, generated using spread.gl (Li et al. 2024), is available as an HTML file Zenodo https://doi.org/10.5281/zenodo.13885992 in Supplementary File 5.

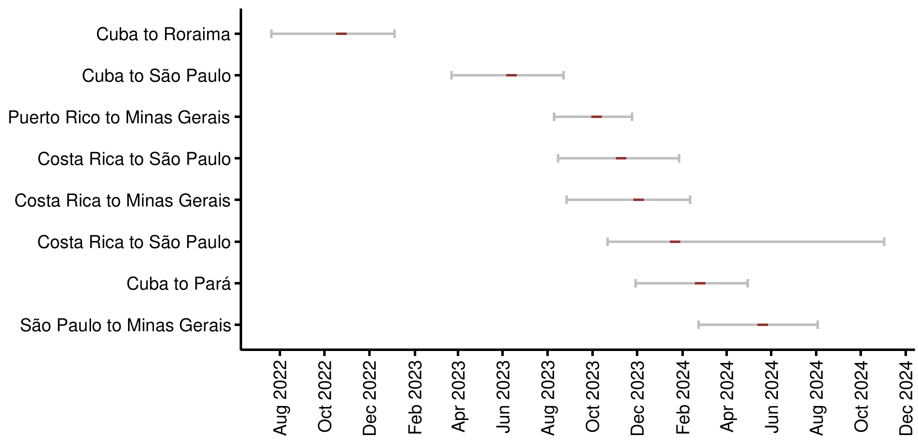

Figure 3. Temporal dynamics of the introductions and spread of minor lineage 3III_B.3.2 in Brazil. The central points represents temporal estimates for the most recent common ancestor of each introduction, while the associated uncertainty (95% HPD interval) is shown by horizontal error bars.

with population susceptibility, may have contributed to the establishment and circulation of this lineage. Nevertheless, the interaction between these factors and their impact on viral dynamics remains uncertain. Further studies are essential to elucidate the mechanisms driving changes in viral lineage composition, as these alterations directly influence outbreak dynamics and DENV transmission patterns in the country (Adams et al. 2006, Nunes et al. 2014, Lourenço et al. 2018).

Despite the lack of genomic data from Brazil between 2010 and 2022, the detection of a limited number of DENV3 cases by serology from 2014 to 2022 suggests the residual circulation of major lineage 3III_C. This hypothesis is supported by the increase in cases observed in 2023, which coincides with our phylogeographic data indicating the introduction of the 3III_B.3.2 lineage into Brazil around October 2022, originating from the Caribbean region. This finding aligns with a previous study suggesting that this lineage was introduced into the Caribbean at the end of 2020, with subsequent spread to Suriname, the USA, and eventually Brazil (Naveca et al. 2023, Taylor-Salmon et al. 2024). These introductions underscore the critical role of international dissemination in the resurgence of DENV-3, particularly in Northern Brazil.

Our analyses indicate that, between 2022 and 2024, multiple independent introductions from the Caribbean region contributed to the resurgence of DENV-3 in various parts of Brazil. Although the number of cases had been relatively low in previous years, the virus re-emerged in 2023, with the highest incidence rates concentrated in the northern states (Ministério da Saúde 2024). The phylogeography suggests that the 3III_B.3.2 lineage was introduced into the state of Minas Gerais through at least two international events with the first occurring in early October 2023, likely originating from the Caribbean region, specifically Puerto Rico, as suggested by previous studies (Naveca et al. 2023, Adelino et al. 2024). In the state of São Paulo, at least three independent introductions were identified, with the first occurring in late June 2023 from Cuba. The most recent international introduction in our analyses was detected in the state of Pará, also originating from Cuba, around early February 2024.

Interestingly, we identified only a single instance of interstate transmission occurring in the Southeastern region of Brazil, between the states of São Paulo and Minas Gerais, likely occurring in mid-May 2024, suggesting that external introductions, rather than domestic spread, were the primary drivers of DENV-3 circulation during this period.

Nine of the new DENV-3 genomes were collected in Minas Gerais between epidemiological weeks 4 and 44 of 2024, indicating the sustained circulation of this serotype. The Southeast region was the most affected in the country, reporting the highest number of DENV-3 cases between 2023 and 2024, with a total of 1187 infections. The state of Minas Gerais, located in this region, accounted for 382 of these cases in 2024 alone, representing the highest number of cases in the past decade, according to data from the National System of Notifiable Diseases (SINAN) ([TabNet, http://tabnet.datasus.gov.br/cgi/deftohtm.exe?sinannet/cnv/denguebbr.def], accessed on 26 February 2025), which is also available on Zenodo ([https://doi.org/10.5281/zenodo.13885992] in Supplementary File 6).

Additionally, two of the new DENV-3 genomes generated in this study represent some of the first documented cases in the state of Pará, located in the northern region of the country. These genomes were collected on 14 May and 9 June 2024, from symptomatic individuals in a rural area of Altamira. The northern region of the country was the most affected by this serotype in 2023 and the second in 2024, reporting 381 cases, 33 of which were reported in the state of Pará in the past year, following a period of absence of recorded circulation of this serotype since 2019, according to SINAN data (Supplementary File 6).

Furthermore, given the notable expansion of DENV transmission in Southern Brazil, particularly in the states of Santa Catarina, Rio Grande do Sul, and Paraná, with 119 804 confirmed cases in 2024—driven by factors such as increased environmental suitability for *Aedes*, a susceptible population, and multiple independent viral introductions (Codeco et al. 2022)—along with the sharp increase in reported DENV-3 cases in the same year (169 cases) (Supplementary File 6), we infer that additional international introductions may have occurred. Furthermore, other domestic transmission routes may also play a significant role in the dissemination of the 3III_B.3.2 lineage in Brazil. However, the limited representation of this region in the dataset constrains a more precise assessment of these dynamics and underscores the need for studies encompassing a broader geographical context.

Limitations related to the incompleteness of the genomes generated in this study, the geographic representativeness of the analysed data—restricted to four Brazilian states—and the limited availability of genomic data from other states in public databases (see Acknowledgements) may affect the generalizability of this study's findings. Additionally, these limitations, combined with case underreporting, may introduce biases regarding the estimated timing of viral introductions. Therefore, we emphasize that extrapolating our conclusions to other regions of Brazil should be done with caution and reinforce the importance of expanding collaborative genomic surveillance efforts to achieve a more representative dataset and more robust analyses.

Conclusion

Our study underscores the re-emergence of DENV-3 in Brazil, driven predominantly by the introduction and circulation of the 3III_B.3.2 lineage. Multiple independent introductions from the Caribbean region, followed by subsequent dispersal across Brazil, highlight the dynamic patterns of viral migration. We acknowledge that the limitations of this study may influence the results and their interpretation. Nevertheless, they also emphasize the necessity for enhanced genomic surveillance and coordinated public health interventions to mitigate the potential for the emergence of widespread DENV-3 outbreaks, particularly in regions with limited prior exposure to this serotype. Ongoing surveillance efforts will therefore be critical in identifying emerging DENV lineages and guiding timely responses to prevent future DENV epidemics in Brazil.

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Author contributions

J.S.P. and A.R.J.L. conceptualized the study and curated the data, as well as performed the formal analyses. S.C.S. and M.C.E. acquired the funding. J.S.P., G.R., G.A.L.B., K.C.L.F., P.M.S.S.B.C., S.M.S.C., T.E.R.A., F.Cd.M.I., M.G., and A.R.J.L. conducted the investigation. J.S.P., I.C.B., V.C.S., I.S.R., I.T.T.L., V.H., M.G., and A.R.J.L. developed the methodology. A.R.J.L. managed the project, while G.A.L.B., K.C.L.F., P.M.S.S.B.C., S.M.S.C., T.E.R.A., L.C.J.A., and F.Cd.M.I. provided resources. J.S.P. was responsible for data visualization. J.S.P., S.N.S., and A.R.J.L. wrote the original draft, and J.S.P., S.N.S., I.C.B., G.R., V.C.S., V.H., N.D.G., S.C.S., L.C.J.A., M.C.E., M.G., and A.R.J.L. reviewed and edited the manuscript.

Supplementary data

Supplementary data is available at *VEVOLU Journal* online.

Conflict of interest: The authors declare no conflicts of interest related to this study.

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Data availability

The dataset of sequences retrieved from GISAID can be accessed using the EPI_SET ID: EPI_SET_250604wy (https://doi.org/10.55876/gis8.250604wy). Additionally, all sequence accession numbers, associated metadata used in this study, BEAST analysis logs, publicly available serological data, and an .html file illustrating the introduction routes are available on Zenodo: https://doi.org/10.5281/zenodo.13885992.

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